An Integrated Data-Mining and Explainable Machine-Learning Framework for Food-Waste Prediction and Risk Analysis

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Abstract

Food-service operations must balance meal availability against overproduction, yet waste-prediction studies often emphasize a single modeling task. This paper documents an integrated analytical workflow combining exploratory and OLAP-style aggregation, K-Means clustering, Apriori association-rule mining, supervised regression, feature ablation, outlier sensitivity analysis, chronological validation, and SHAP explanations. The notebook reports 911 observations across 867 unique dates from January 1, 2022, to September 26, 2024. Four regressors were compared using three expanding-window validation folds within a development period, followed by a held-out future-period test. Random Forest was selected by mean validation RMSE (10.262 kg) and achieved test MAE of 5.315 kg, RMSE of 8.505 kg, and R^2 of 0.894 on 174 test observations. SHAP ranked meals served and the supplied past-waste variable as the leading contributors to predictions. The clustering result was highly imbalanced, while feature ablation and outlier sensitivity indicated strong dependence on meal volume and unusual operating cases. The workflow is therefore promising as an exploratory decision-support prototype, not proof of real-world waste reduction. Because the input CSV and its provenance were not included with the materials reviewed, the dataset's origin and the reported results could not be independently verified; this limitation must be resolved before publication.

Index Terms—*food-waste prediction, machine learning, Random Forest, association-rule mining, clustering, temporal validation, SHAP, decision support.*

I. INTRODUCTION

Food waste has environmental, economic, and social consequences, and food-service settings present a specific planning challenge: kitchens must prepare enough food to meet demand without consistently producing more than is consumed. Forecasting work in public catering has shown that better estimates of attendance or meal demand can support portion planning, while still requiring safeguards against shortages [1], [2]. These studies focus mainly on demand or attendance; predicting the amount of waste directly is a related but distinct task because waste can also depend on preparation practices, staffing, environmental conditions, events, and the recent history of waste.

Data mining and machine learning offer tools for analyzing such operational records. Regression models can estimate a continuous waste quantity, association rules can summarize co-occurring conditions, clustering can identify unusual operating regimes, and explainable AI can expose which variables influence model predictions [5]–[8]. However, prediction quality alone is not sufficient for practical use. Evaluation must respect time ordering, explanatory patterns must not be mistaken for causal effects, and extreme observations must be examined rather than removed automatically.

This study develops and documents an applied workflow around a food-waste analysis notebook. The workflow uses operational, environmental, calendar, event, and historical-waste variables; compares Linear Regression, Random Forest, Gradient Boosting, and XGBoost; and evaluates the selected model on a future, date-separated test period. It additionally examines feature ablation, predictor-extreme observations, association rules, clustering, and SHAP values. The goal is to assess what the supplied experiment demonstrates, while identifying the checks required before the workflow can be considered a validated operational system.

The contribution is methodological integration rather than a new learning algorithm. The manuscript reports the notebook's saved outputs, gives priority to a date-aware evaluation design, and interprets weaknesses alongside the headline metrics. It does not claim that the system has already reduced food waste in practice. A critical limitation is that the notebook references food_waste_cleaned.csv, but that data file and its collection documentation were not supplied with the artifacts reviewed for this manuscript. Consequently, the data's provenance and the real-world meaning of the recorded values remain unverified.

II. RELATED WORK

Malefors et al. studied guest-attendance forecasting in 21 Swedish public-catering kitchens using historical portion counts from 2010–2019. Their comparison included simple last-value and moving-average methods as well as Prophet and neural-network models; forecasts were connected to portion-planning decisions [1]. The study illustrates both the value of temporal evaluation and the importance of balancing forecast error against the risk of underproduction.

Rodrigues et al. compared machine-learning approaches for short-term demand forecasting using data from three catering services. Their methods included Random Forest and long short-term memory models, with baseline comparisons. The authors reported estimated reductions in wasted meals when forecasts were used for meal planning, while also considering unmet demand [2]. These results concern demand forecasting and operational decision simulations in those case studies; they should not be interpreted as an effect established by the present notebook.

Shadid et al. reviewed 77 peer-reviewed studies on data-driven food-waste and food-loss management and identified a fragmented research landscape, with continuing challenges involving data access, scalability, and integration across contexts [3]. The review motivates workflows that connect descriptive analysis, prediction, interpretation, and implementation, but integration by itself does not prove novelty or generalizability.

Ghuriani et al. applied machine-learning classifiers and explainable AI to a survey of 341 respondents in India, focusing on household food-waste behavior and behavioral attributes [4]. Their domain and outcome differ from the operational regression task examined here, yet the work is relevant in showing the value of interpreting predictions rather than presenting accuracy alone. Random Forests and gradient-boosted trees are established methods [5], [6], while SHAP provides a unified framework for feature attribution [7]. Apriori-style association-rule mining is a standard technique for discovering frequent co-occurrences [8].

The present work therefore addresses an applied workflow question: can a single notebook combine time-aware direct waste regression with exploratory pattern mining and model explanations, and what limitations become visible when those outputs are considered together? The paper does not claim that no previous study has combined these methods; a comprehensive systematic review and a larger comparison of published systems would be required to support that stronger novelty claim.

III. RESEARCH QUESTIONS AND CONTRIBUTIONS

The experiment is organized around three questions. RQ1: How do four conventional regression models compare when the development and test sets are separated chronologically? RQ2: Which predictor groups and features appear most influential, and how sensitive is validation performance to removing predictor-extreme training cases? RQ3: What patterns are surfaced by clustering, OLAP-style aggregation, and association-rule mining, and how cautiously should those patterns be interpreted?

The main deliverable is an end-to-end applied workflow with a separate future-period test, alongside a set of diagnostic analyses that make model dependencies visible. The workflow is reproducible in principle from the notebook and input data, but full independent reproduction is currently blocked by the absence of the CSV and source documentation. The study makes no claim of algorithmic novelty, causal inference, or measured waste reduction.

IV. DATASET AND PREPROCESSING

A. Dataset Overview

The saved notebook output reports 911 rows and 21 input columns at load time, followed by engineered temporal and descriptive fields. The observed date range is January 1, 2022, through September 26, 2024, with 867 unique dates. Because there are more rows than unique dates, some dates contain multiple observations. The target is food_waste_kg. Listed variables include meals_served, kitchen_staff, temperature_c, humidity_percent, day_of_week, special_event, past_waste_kg, staff_experience, waste_category, and date. The modeling matrix uses 13 predictor columns: meals_served, kitchen_staff, temperature_c, humidity_percent, day_of_week, special_event, past_waste_kg, year, month, day, day_of_week_num, week_of_year, and is_weekend.

The input file used by the code is named food_waste_cleaned.csv. That file was not included with the notebook and manuscript artifacts available for this authoring task. Thus the dataset cannot be independently rerun here, and it is not possible to confirm whether the records are directly measured, synthetically generated, transformed from another source, or otherwise curated. The following numerical findings are consequently identified as notebook-reported results rather than independently reproduced observations.

TABLE I. DESCRIPTIVE STATISTICS REPORTED BY THE NOTEBOOK

Variable	Mean ± SD	Range	Median
Meals served	375.41 ± 502.81	100–4730	306.00
Kitchen staff	11.90 ± 4.29	5–19	12.00
Temperature (°C)	22.19 ± 8.92	−10.37–60.00	22.12
Humidity (%)	60.76 ± 17.33	30.12–89.98	61.63
Past waste (kg)	27.02 ± 12.77	5.01–49.80	26.85
Target waste (kg)	44.84 ± 27.93	10.82–274.33	41.15
Waste per meal (kg)	0.150 ± 0.064	0.030–0.476	0.137

The target distribution is right-skewed: most observations lie in the lower tens of kilograms, while a small number of observations exceed 150 kg. The reported maximum meal count of 4,730 and temperature maximum of 60 °C deserve source-level inspection. They may be valid unusual records, unit or measurement problems, or artifacts of data generation; summary statistics alone cannot determine which. The analysis retains these cases and separately measures sensitivity to removing predictor-extreme training rows.

B. Cleaning and Feature Engineering

The notebook standardizes column names, parses dates with invalid-date coercion, removes exact duplicate rows, converts selected fields to numeric types, and removes rows with missing targets. The saved output reports zero missing values in the final analysis dataframe and no duplicate rows removed. Calendar features are derived from date: year, month, day, numeric day of week, ISO week of year, and a weekend flag. Waste per meal and waste per 100 meals are calculated for descriptive analysis but excluded from the model predictors because they directly contain the target quantity. Categorical encoding and imputers are placed inside the modeling pipeline. Categorical fields are one-hot encoded with unknown-category handling; numeric values pass through a median-imputation step. Since the reported modeling columns contain no missing values, the imputation steps did not need to fill missing values in the saved run.

One semantic issue requires confirmation: past_waste_kg is treated as a historical predictor by the notebook, but the input data and code do not establish how that field was produced. Before making a forecasting claim, the researcher should verify that this field was available at the intended prediction time and does not encode current or future target information. Duplicate dates are kept within a single split to avoid same-date overlap between development and test sets.

V. METHODOLOGY

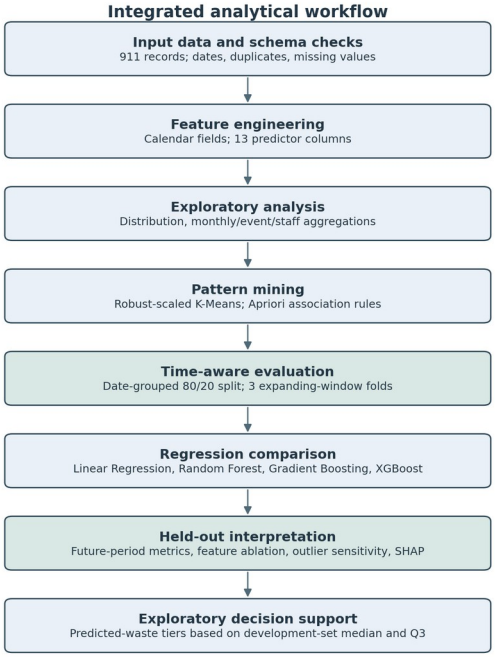


Fig. 1. Workflow represented by the supplied notebook. The final stage is an exploratory risk-tier output, not a measured intervention.

A. Exploratory and OLAP-Style Analysis

Descriptive summaries and grouped aggregations compare total and average waste by year and month, by special-event status, and by kitchen-staff count. The saved output reports 833 non-event observations with mean waste of 43.83 kg and 78 event observations with mean waste of 55.68 kg. Mean meal counts are similar between these two groups (375.46 and 374.77, respectively), but these unadjusted aggregates do not establish a causal event effect. The notebook implements these summaries as dataframe aggregations. It does not provide evidence of a deployed relational data warehouse or a physically instantiated star schema, so the analysis is described as OLAP-style rather than as a validated warehouse implementation.

B. K-Means Clustering

K-Means clustering [10] is applied after RobustScaler transformation to meals served, kitchen staff, temperature, humidity, past waste, and current waste. Candidate cluster counts from two through six are compared by silhouette coefficient. Two clusters receive the highest reported silhouette value (0.8825). The large cluster contains 891 observations and averages 305.95 meals and 41.28 kg of waste; the smaller cluster contains 20 observations and

averages 3,469.50 meals and 203.39 kg of waste. Because current waste is included in the clustering inputs and the resulting groups are highly imbalanced, the result is more appropriately interpreted as separation of an extreme-volume regime than as proof of two stable operational types. A future version should also cluster predictors without the target and test stability across resamples.

TABLE II. REPORTED CLUSTER PROFILES

Cluster	Records	Mean meals	Mean waste (kg)
0	891	305.95	41.28
1	20	3469.50	203.39

C. Association-Rule Mining

Apriori is applied to a transaction-encoded dataset of discretized conditions. A rule A → B is summarized by support P(A ∩ B), confidence P(B|A), and lift P(B|A)/P(B) [8]. The notebook reports frequent rules whose consequent is high waste. The strongest shown combination is high meals, high past waste, and high staff, with support 0.116, confidence 0.946, and lift 1.891. The simpler high-meals plus high-past-waste rule has support 0.233, confidence 0.926, and lift 1.850. A special-event rule has support 0.069, confidence 0.808, and lift 1.614. These statistics describe co-occurrence in the encoded data; they neither demonstrate causal effects nor generalize automatically to another kitchen.

TABLE III. EXAMPLES OF HIGH-WASTE ASSOCIATION RULES

Antecedent → High waste	Support	Conf.	Lift
High meals + high past waste + high staff	0.116	0.946	1.891
High meals + high past waste	0.233	0.926	1.850
Special event = yes	0.069	0.808	1.614
High past waste	0.371	0.741	1.481

D. Regression Models

The continuous target food_waste_kg is predicted using four regressors: Linear Regression, Random Forest [5], Gradient Boosting, and XGBoost [6]. Random Forest uses 300 trees and random_state=42. Gradient Boosting uses 300 estimators, a learning rate of 0.05, depth 3, and random_state=42. XGBoost uses 300 estimators, learning rate 0.05, maximum depth 4, subsample 0.8, column subsampling 0.8, squared-error loss, and random_state=42. Linear Regression is retained as a simple reference model. A ColumnTransformer handles numeric and categorical features inside a pipeline, and the same preprocessing logic is fitted independently within each fold.

E. Chronological Model Selection and Final Test

Rows are sorted by date and split by unique dates, not by row-level random sampling. The first 80% of unique dates form the development set (737 rows) and the final 20% form the future test set (174 rows). The development period runs from January 1, 2022, to March 10, 2024; the test period runs from March 11 to September 26, 2024. The notebook confirms there are no dates shared between the two sets. Within development, three expanding-window folds train on the earlier 50%, 65%, and 80% of unique development dates and validate on the next 15%, 15%, and 20%, respectively. Model selection uses the lowest mean RMSE across these three chronological validation folds. After selection, the chosen model is refit on all development observations and evaluated once on the future test set.

For observed targets y_i and predictions $\hat{y}_i$ over n observations, the three reported metrics are defined as follows:

$$MAE = (1/n)\,\Sigma_i\,|y_i - \hat{y}_i| \quad (1)$$

$$RMSE = \sqrt{[(1/n)\,\Sigma_i\,(y_i - \hat{y}_i)^2]} \quad (2)$$

$$R^2 = 1 - \Sigma_i(y_i - \hat{y}_i)^2 / \Sigma_i(y_i - \bar{y})^2 \quad (3)$$

MAE represents average absolute error in kilograms; RMSE penalizes large errors more strongly; R^2 compares residual error with variability around the observed mean. Validation R^2 values are averaged across folds and should not be compared directly with the single final-test R^2 without accounting for the different evaluation blocks.

F. Feature Ablation and Outlier Sensitivity

Feature ablation is evaluated on the final chronological validation block within the development set, rather than on the final future test set. Four feature subsets are compared: all predictors, meals plus past waste, meals only, and all predictors except meals. Outlier sensitivity uses the same development-period validation block. A union of IQR-based flags across meals served, kitchen staff, temperature, humidity, and past waste is calculated from training data only; 30 predictor-extreme observations are removed from the 598-row training subset, leaving 568 observations. The purpose is sensitivity analysis, not a presumption that unusual observations are erroneous.

G. SHAP and Risk-Tier Output

SHAP explanations are computed using a model-agnostic explainer around the final pipeline. A random sample of up to 100 development records is used as the background and up to 200 final-test records are sampled for explanation; because the final test contains 174 observations, all 174 can be explained. Mean absolute SHAP values are used to rank global feature influence [7]. Risk categories are assigned from predicted waste using the development-target median and 75th percentile: Low at or below the median, Medium above the median and at or below the 75th percentile, and High above the 75th percentile. These categories are relative to the dataset distribution, not calibrated safety thresholds and not validated operational recommendations.

VI. RESULTS

A. Time-Aware Model Comparison

Random Forest had the lowest mean validation RMSE among the four candidates, at 10.262 kg, and was selected under the predefined selection rule. XGBoost had a slightly higher mean MAE/RMSE profile but the highest mean fold R^2 (0.757). Thus, Random Forest was chosen specifically for lower mean RMSE, not because it dominated every metric. This distinction matters because the fold-to-fold metrics vary: the first fold includes a notably lower Random Forest R^2 than its later folds.

TABLE IV. MEAN METRICS OVER THREE CHRONOLOGICAL VALIDATION FOLDS

Model	MAE (kg)	RMSE (kg)	Mean R^2
Random Forest	5.761	10.262	0.740
XGBoost	5.922	11.716	0.757
Gradient Boosting	6.093	12.146	0.604
Linear Regression	6.572	12.319	0.740

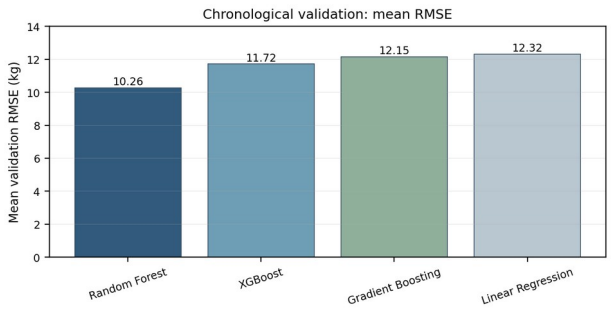


Fig. 2. Mean RMSE across the notebook's three expanding-window validation folds.

B. Future-Period Test Performance

The selected Random Forest model achieved test MAE 5.315 kg, RMSE 8.505 kg, and R^2 0.8935 across 174 observations from March 11 to September 26, 2024. Relative to the development-fold mean RMSE, the test RMSE is lower; this is not inherently contradictory because the final test period may have a different distribution and contains a limited number of extreme observations. The reported values indicate good fit on this particular holdout, but they do not establish performance across other institutions, seasons, or datasets. There is no external-site validation and no baseline persistence or seasonal forecasting model in the supplied code.

TABLE V. FINAL HELD-OUT FUTURE-PERIOD RESULT

Model	Test MAE	Test RMSE	Test R^2
Random Forest	5.315 kg	8.505 kg	0.894

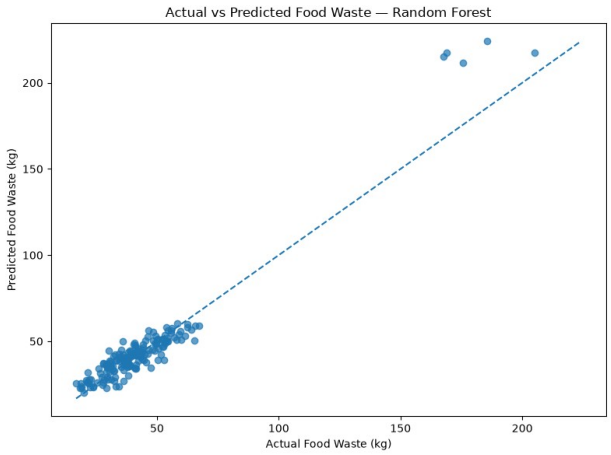


Fig. 3. Actual versus predicted food waste on the final chronological test period. The high-value cases remain visible.

Visual inspection of the saved actual-versus-predicted and residual plots shows that most observations cluster in a lower waste range, while a few extreme predictions are associated with large residuals. For example, one saved test row has actual waste of approximately 168.9 kg and predicted waste of approximately 217.4 kg, an absolute error near 48.5 kg. Aggregate MAE can conceal these high-consequence misses; reporting error stratified by waste level or operational regime is therefore important.

C. Feature Ablation and Outlier Sensitivity

TABLE VI. FEATURE ABLATION ON THE INNER DEVELOPMENT VALIDATION BLOCK

Feature set	MAE (kg)	RMSE (kg)	R^2
All 13 predictors	5.900	9.781	0.919
Meals + past waste	7.422	12.583	0.866
Meals only	11.543	16.911	0.759
All except meals	14.762	33.613	0.047

The inner development validation block indicates that all features perform best among the tested ablations. The meals-plus-past-waste subset is materially weaker but retains useful signal, while removing meals causes a large increase in RMSE and a drop in R^2 to 0.047. This supports the interpretation that meal volume is a key predictor in the saved data. However, this comparison reflects one validation block and was not repeated over all folds; it should be treated as a diagnostic rather than a stable estimate of feature-group effects.

TABLE VII. SENSITIVITY TO REMOVING IQR-EXTREME TRAINING ROWS

Training condition	n	MAE	RMSE	R^2
Original training rows	598	5.900	9.781	0.919
30 predictor-extreme rows removed	568	10.626	32.433	0.112

Removing 30 predictor-extreme records from training substantially worsens validation RMSE from 9.781 to 32.433 kg. This result argues against automatic IQR-based deletion without verifying that a record is erroneous. Some unusual meal-volume or operational cases may be needed for the model to learn rare conditions. At the same time, the sensitivity result is confounded by a changed training sample and a small number of very high-waste validation cases; it does not prove that all flagged observations are valid.

D. Association Rules and Clustering

The highest-confidence association in the notebook is high meals + high past waste + high staff → high waste (confidence 0.946, lift 1.891). The pair high meals + high past waste has support 0.233, confidence 0.926, and lift 1.850. These values are consistent with the model's reliance on meals served and the historical-waste field. However, the categorical cut points and rule generation were exploratory, and the notebook does not include a separate future-period replication of the rules. A stronger study would estimate cutoffs and patterns using only development data, then evaluate their stability on later dates.

The silhouette score of 0.883 for K=2 is high in isolation, but the cluster profiles reveal that the small cluster is dominated by 20 observations with more than 3,000 meals on average. This imbalance suggests that the clustering is separating extreme scale from typical scale. Since the target food waste was also included in the clustering variables, the resulting partition is not an independent validation of operational regimes. The cluster analysis should therefore be reported as exploratory anomaly/regime discovery, not as robust segmentation.

E. SHAP Interpretation and Risk Categories

TABLE VIII. TOP FIVE MEAN ABSOLUTE SHAP VALUES

Rank	Feature	Mean |SHAP|
1	Meals served	10.063 kg
2	Past waste	6.816 kg
3	Kitchen staff	1.078 kg
4	Special event	1.072 kg
5	Humidity	0.392 kg

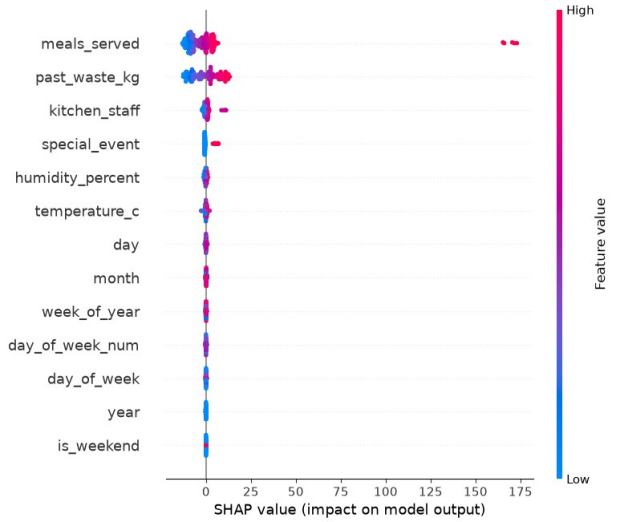


Fig. 4. SHAP summary plot recorded by the notebook for the final Random Forest pipeline.

The global SHAP ranking identifies meals_served (mean absolute SHAP 10.063 kg) and past_waste_kg (6.816 kg) as the two dominant contributors, with kitchen_staff and special_event a distant third and fourth. This is consistent with the ablation result showing a sharp deterioration when meals_served is excluded. SHAP expresses contribution to the model output, not causal importance: the values do

not prove that changing the number of meals or staffing will directly change waste by that amount. The risk-tier summary separates predictions into Low, Medium, and High groups using the development median and upper quartile. This improves communication of model output but does not demonstrate that the categories reduce waste or support an optimized preparation policy.

VII. DISCUSSION

Three practical observations emerge from the saved experiment. First, meal volume is central to prediction in this dataset. Both ablation and SHAP rank it highly, and removing the feature creates the largest deterioration among the tested subsets. This is plausible operationally because the amount of food served can scale with the amount of waste; however, that same relationship makes the prediction timing important. If the intended use is to decide how much food to prepare before service, actual meals_served may not yet be available. A deployable forecast would then need planned attendance or a separately forecast demand input rather than the realized count.

Second, the past_waste_kg feature adds substantial predictive information, but its derivation is not documented in the provided notebook. The author should confirm whether it is a true lagged measurement and whether every such value was available at the time predictions were made. If it was computed using the target, inferred from a formula, or generated after the observation, the evaluation may contain target leakage or data-generation artifacts. This question is more consequential than further hyperparameter tuning.

Third, aggregate metrics should be read alongside errors on rare cases. The model obtains a high held-out R^2, but visual diagnostics show conspicuous high-value errors. Given the strongly right-skewed target, future evaluation should include MAE/RMSE by quantile, performance on the top 5–10% of waste observations, and a robust baseline. A simple previous-observation predictor and a weekday seasonal naive predictor would clarify whether the learned models add value over cheap operational heuristics. The existing paper also does not estimate any financial savings or physical waste reduction, so those outcomes should not be claimed.

The combined workflow can be useful as a decision-support prototype because different methods provide different perspectives: temporal aggregation shows seasonal variation; association rules summarize recurring combinations; clustering highlights a high-volume regime; regression produces numeric estimates; SHAP provides model explanations. But an integrated dashboard or notebook becomes a research contribution only when each analysis serves a defined question and its reliability is tested. Some methods may be removed from a shorter paper if they do not add validated insight beyond the main regression task.

VIII. LIMITATIONS AND VALIDITY THREATS

Dataset provenance and reproducibility: The CSV used by the notebook was not supplied in the materials reviewed for this manuscript, and there is no collection protocol or external citation for it. It is therefore impossible to independently confirm whether the observations are measured operational records, simulated data, or transformed data from another source. The notebook's saved outputs are reported faithfully, but the results should not be presented as validated real-world evidence until the file, provenance, license, and data dictionary are documented.

Potential feature-timing or target leakage: The input named past_waste_kg is treated as historical waste, but the notebook does not generate this column from a sorted target lag or document its upstream source. The model's strong reliance on this variable makes a data-lineage audit essential. Likewise, meals_served may be post-event information rather than an available input for pre-service decisions. The precise prediction time must be defined before deployment.

Limited evaluation: There is one chronological holdout and three expanding-window folds, but no cross-site test, repeated temporal backtest, persistence baseline, seasonal naive baseline, uncertainty interval, or statistical comparison. The ablation and outlier experiments use a single inner validation block. The high R^2 should not be generalized beyond this split.

Exploratory mining: Current waste is included in K-Means, the two-cluster solution is heavily imbalanced, and association rules are not independently validated in a later period. Association, clusters, SHAP values, and feature importance are descriptive/model-based outputs and do not establish causation. Risk thresholds based on median and upper quartile are distribution-relative rather than domain-defined operational thresholds.

Model and implementation scope: The provided artifact is a Jupyter notebook. It does not include the code for a deployed Streamlit application or a physically implemented data warehouse, despite the broader decision-support ambition in the older draft. Such components should only be claimed after the implementation is provided and tested. The experiments also do not show that use of the model reduces food waste, cost, or unmet demand.

IX. CONCLUSION AND FUTURE WORK

This paper documented an applied food-waste analysis workflow integrating OLAP-style aggregation, K-Means, Apriori association rules, four regression models, chronological model selection, an untouched future-period test, feature ablation, predictor-outlier sensitivity, SHAP explanations, and relative risk labels. On the saved notebook outputs, Random Forest achieved a final-test MAE of 5.315 kg, RMSE of 8.505 kg, and R^2 of 0.894. Meals served and the supplied past-waste variable dominated model explanations. At the same time, the clustering separated a small extreme-volume group, unusual observations had a strong effect on error, and the largest test errors occurred among high-waste cases.

The work is best regarded as a promising exploratory prototype, not yet a validated food-service forecasting system. The most important next steps are to publish or document the dataset source, verify that historical predictors are truly available before prediction, add persistence and seasonal baselines, evaluate errors on high-waste cases, repeat rolling-origin tests, and validate on an independent kitchen or institution. If the intended benefit is reduced preparation waste, a prospective operational study must measure waste, shortage risk, and costs before and after adoption. Until these checks are complete, the results should be interpreted as notebook-reported evidence on one dataset rather than proof of generalizable performance or impact.

DATA AND CODE AVAILABILITY

The submitted notebook expects an input file named food_waste_cleaned.csv and writes tables and figures to a paper_results directory. The CSV was not included with the files available for preparing this manuscript, and no dataset source or license was provided. Before submission, the author should add a stable dataset citation or archive the data where permitted, document how past_waste_kg is defined, and verify the notebook outputs from the archived input. The notebook itself is the implementation basis for the analysis described here.

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